

Mark Scheme — Practice Paper 5 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 $(x + \frac{5}{2})^2 - \frac{33}{4}$ (B1 $a = \frac{5}{2}$, B1 $b = -\frac{33}{4}$). Minimum value $-\frac{33}{4} = -8.25$ (B1 FT). [3]

2 $2(1 - \cos^2 \theta) + 3 \cos \theta = 0 \Rightarrow 2 \cos^2 \theta - 3 \cos \theta - 2 = 0$ (M1 A1). $(2 \cos \theta + 1)(\cos \theta - 2) = 0 \Rightarrow \cos \theta = -\frac{1}{2}$ (M1). $\theta = 120^\circ, 240^\circ$ (A1). [4]

3(a) $1 + 10x + 40x^2 + 80x^3$ (B1 for $1 + 10x$, B1 for $40x^2 + 80x^3$). [2]

3(b) $1 \times 80 + (-3) \times 40$ (M1) $= -40$ (A1). [2]

4(a) $\frac{a}{1-r} = 3a \Rightarrow 1-r = \frac{1}{3}$ (M1) $\Rightarrow r = \frac{2}{3}$ (A1). [2]

4(b) $a + \frac{2}{3}a = 20 \Rightarrow a = 12$ (B1). $S_6 = \frac{12(1 - (2/3)^6)}{1 - 2/3}$ (M1) $= 32.8$ (A1; 32.84). [3]

5(a) $\frac{1}{2}r^2\theta = 24$, $r\theta = 8$ (B1). Dividing: $\frac{1}{2}r = 3 \Rightarrow r = 6$ (M1), $\theta = \frac{4}{3}$ (A1). [3]

5(b) Segment $= 24 - \frac{1}{2}(6)^2 \sin \frac{4}{3}$ (M1 A1) $= 24 - 17.50 = 6.50 \text{ cm}^2$ (A1). [3]

6(a) $P(8, 0)$, $Q(0, 6)$ (B1). $PQ = \sqrt{64 + 36} = 10$ (B1). [2]

6(b) Centre = midpoint $(4, 3)$ (B1), radius = 5 (B1). $(x - 4)^2 + (y - 3)^2 = 25$ (A1). At O : $16 + 9 = 25$ ✓, so O lies on the circle (B1). [Equivalently, angle $POQ = 90^\circ$ is the angle in a semicircle.] [4]

7(a) $x = 5 - 2y \Rightarrow y = \frac{5-x}{2}$ (M1); $f^{-1}(x) = \frac{5-x}{2}$ (A1). [2]

7(b) $fg(x) = 5 - 2(x^2 + 1) = 3 - 2x^2$ (B1). Range: $fg(x) \leq 3$ (B1). [2]

7(c) $\frac{5-x}{2} = x^2 + 1 \Rightarrow 2x^2 + x - 3 = 0$ (M1 A1). $(2x + 3)(x - 1) = 0 \Rightarrow x = 1$ only, since $x \geq 0$ (A1). [3]

8(a) $\frac{dy}{dx} = 2x - \frac{16}{x^2}$ (B1). $= 0 \Rightarrow x^3 = 8 \Rightarrow x = 2$, $y = 4 + 8 = 12$ (M1 A1). $\frac{d^2y}{dx^2} = 2 + \frac{32}{x^3} = 6 > 0$, minimum (A1). [4]

8(b) At $x = 4$: $\frac{dy}{dx} = 8 - 1 = 7$ (B1). $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = 7 \times 0.05$ (M1) $= 0.35$ units per second (A1). [3]

9(a) $\cos^4 \theta - \sin^4 \theta = (\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta)$ (M1) $= \cos^2 \theta - \sin^2 \theta$ (M1) $= (1 - \sin^2 \theta) - \sin^2 \theta = 1 - 2 \sin^2 \theta$ (A1 AG). [3]

9(b) $1 - 2 \sin^2 \theta = \frac{1}{2} \Rightarrow \sin^2 \theta = \frac{1}{4}$ (M1) $\Rightarrow \sin \theta = \pm \frac{1}{2}$. In $0^\circ \leq \theta \leq 180^\circ$: $\theta = 30^\circ$ (A1), 150° (A1). [3]

9(c) $2x - 30^\circ = 45^\circ, 225^\circ$ (M1) $\Rightarrow x = 37.5^\circ, 127.5^\circ$ (A1). [2]

10(a) $x^2 = x + 2 \Rightarrow (x - 2)(x + 1) = 0$ (M1). $A(-1, 1)$, $B(2, 4)$ (A1). [2]

10(b) Area $= \int_{-1}^2 (x + 2 - x^2) dx$ (M1) $= [\frac{1}{2}x^2 + 2x - \frac{1}{3}x^3]_{-1}^2$ (A1) $= \frac{10}{3} - (-\frac{7}{6}) = \frac{9}{2} = 4.5$ (A1). [3]

10(c) $V = \pi \int_0^4 x^2 dy = \pi \int_0^4 y dy$ (M1 A1) $= \pi [\frac{1}{2}y^2]_0^4 = 8\pi$ (A1). [3]

11(a)(i) $a + 2d = 11$; $6(2a + 11d) = 300 \Rightarrow 2a + 11d = 50$ (B1 both). Solve: $7d = 28 \Rightarrow d = 4$, $a = 3$ (M1 A1). **[3]**

11(a)(ii) $3 + 4(n - 1) < 200 \Rightarrow n < 50.25$ (M1). 50 terms (A1). **[2]**

11(b) $\frac{k+6}{k} = \frac{4k}{k+6} \Rightarrow (k+6)^2 = 4k^2$ (M1). $k+6 = \pm 2k$ (or $3k^2 - 12k - 36 = 0 \Rightarrow k^2 - 4k - 12 = 0$) (A1). $k = 6$, $r = 2$ (A1); $k = -2$, $r = -2$ (A1). **[4]**

12(a) $(x-3)^2 + (2x+2)^2 = 20 \Rightarrow 5x^2 + 2x - 7 = 0$ (M1 A1). $(5x+7)(x-1) = 0$: $A(1, 3)$, $B(-\frac{7}{5}, -\frac{9}{5})$ (A1). **[3]**

12(b) $M = \left(\frac{1-1.4}{2}, \frac{3-1.8}{2} \right)$ (M1) $= (-0.2, 0.6)$ (A1). **[2]**

12(c) $C(3, -1)$; gradient $CM = \frac{0.6 - (-1)}{-0.2 - 3} = \frac{1.6}{-3.2} = -\frac{1}{2}$ (M1 A1). Gradient of AB is 2, and $2 \times (-\frac{1}{2}) = -1$, so $CM \perp AB$ (A1). **[3]**

Total: 75